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Track: Cloud Platform Development
Course: Terraform
Lab 2

1-create two workspaces dev and prod

```
bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$ terraform workspace new dev

Warning: Deprecated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$
```

```
bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$ terraform workspace new prod

Warning: Deprecated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$
```

```

bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$ terraform workspace list
t

Warning: Depreciated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile"
instead.

default
dev
* prod

bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$

```

2-create two variable definition files(.tfvars) for the two environments

```

Apply complete! Resources: 22 added, 0 changed, 0 destroyed.

Outputs:

environment = "dev"
region = "us-east-1"
vpc_cidr = "10.0.0.0/16"
bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$

```

[Alt+S]
United States (N. Virginia)
Bassant Ali Kamal (707938860152)
Bassant Ali Kamal

Instances (1/2) Info
Last updated 1 minute ago
Connect
Instance state
Actions
Launch instances

Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
dev-app	i-047e82690b03a99dc	Running	t3.micro	3/3 checks passed	View alarms	us-east-1d	-
dev-bastion	i-08082289dabf02c9f	Running	t3.micro	3/3 checks passed	View alarms	us-east-1d	-

i-08082289dabf02c9f (dev-bastion)

Answer private resource DNS name
-

Auto-assigned IP address
18.212.150.141 [Public IP]

IAM role
-

IMDSv2
Required

Operator
-

Instance type
t3.micro

VPC ID
vpc-0d52705451c3cc7de (main-vpc)

Subnet ID
subnet-0b70553ee7ac21020 (public-subnet-1)

Instance ARN
arn:aws:ec2:us-east-1:707938860152:instance/i-08082289dabf02c9f

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name
-

Managed
false

```
Apply complete! Resources: 22 added, 0 changed, 0 destroyed.
```

Outputs:

```
environment = "prod"
region = "eu-north-1"
vpc_cidr = "172.16.0.0/16"
bassant@fedora:~/Documents/Terraform/Terraform_ITI/Labs$
```

[Alt+S] [Europe (Stockholm)] Bassant Ali Kamal (707938860152)

Instances (1/14) Info Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Choose filter set

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
bastion_instance	i-0569f085058639301	Terminated	t3.micro	-	View alarms	eu-north-1b	-
application_in...	i-07bcd8a1e0bf569e3	Terminated	t3.micro	-	View alarms	eu-north-1b	-
☒ prod-bastion	i-06855be38eb876618	Running	t3.small	Initializing	View alarms	eu-north-1a	-
dev-bastion	i-0da549a59ad9b47a9	Terminated	t3.micro	-	View alarms	eu-north-1a	-
prod-app	i-0a4f53a58b30f99d4	Running	t3.small	Initializing	View alarms	eu-north-1a	-
dev-app	i-0c645591b2d6e07e9	Terminated	t3.micro	-	View alarms	eu-north-1a	-
dev-app	i-081a4a7a4f00c7725	Terminated	t3.micro	-	View alarms	eu-north-1a	-

i-06855be38eb876618 (prod-bastion)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary Info

Instance ID i-06855be38eb876618	Public IPv4 address 16.16.28.31 open address	Private IPv4 addresses 10.0.1.184
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-184.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-1-184.eu-north-1.compute.internal	

3-separate network resources into network module

done

4-apply your code to create two environments one in us-east-1 and eu-central-1

done

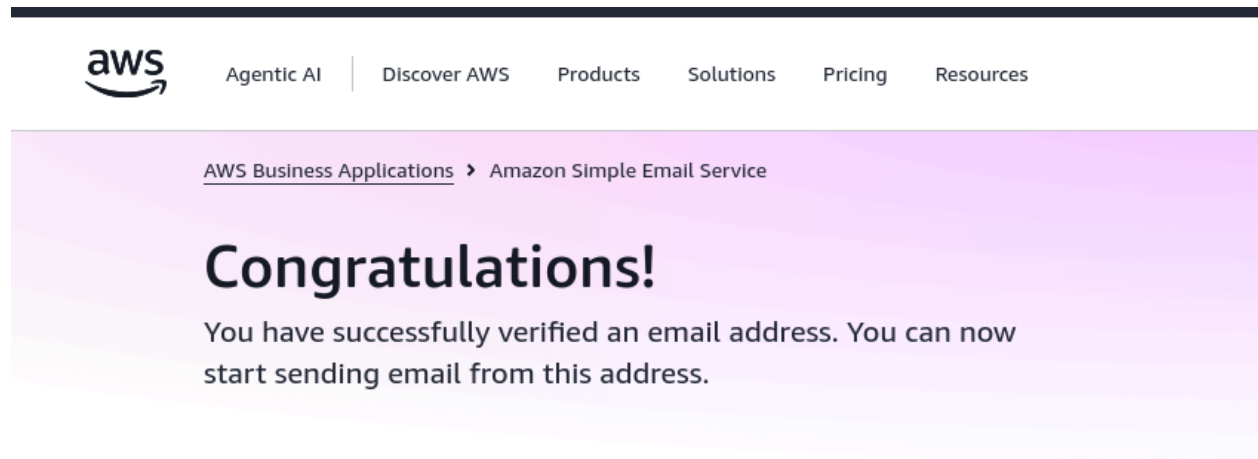
5-run local-exec provisioner to print the public_ip of bastion ec2

```
aws_instance.app: Provisioning with 'local-exec'...
aws_instance.app (local-exec): Executing: ["/bin/sh" "-c" "echo 10.0.3.22"]
aws_instance.app (local-exec): 10.0.3.22
aws_instance.app: Creation complete after 15s [id=i-0ac53e17a7d9909bc]
aws_instance.bastion: Still creating... [00m10s elapsed]
aws_instance.bastion: Provisioning with 'local-exec'...
aws_instance.bastion (local-exec): Executing: ["/bin/sh" "-c" "echo 3.86.25.185"
]
aws_instance.bastion (local-exec): 3.86.25.185
aws_instance.bastion: Creation complete after 14s [id=i-09e9b12e1f4846489]
```

6- upload infrastructure code on github project

done

7-verify your email in ses service



8- create lambda function to send email

lambda-ses-role [Info](#) [Delete](#)

Summary [Edit](#)

Creation date
May 06, 2026, 19:24 (UTC+03:00)

Last activity
-

ARN
[arn:aws:iam::707938860152:role/lambda-ses-role](#)

Maximum session duration
1 hour

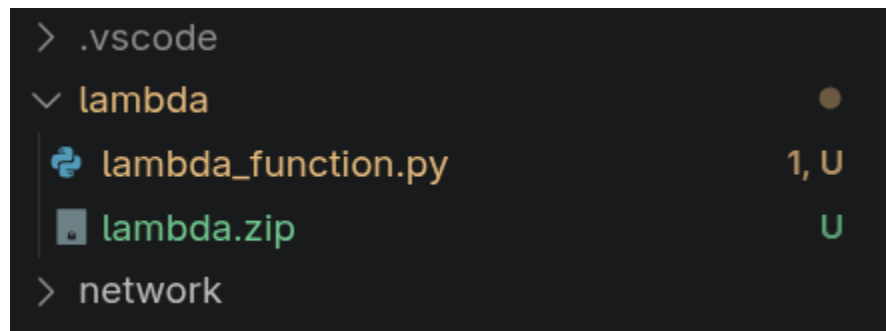
[Permissions](#) | [Trust relationships](#) | [Tags](#) | [Last Accessed](#) | [Revoke sessions](#)

Permissions policies (2) [Info](#) [Simulate](#) [Remove](#) [Add permissions](#)

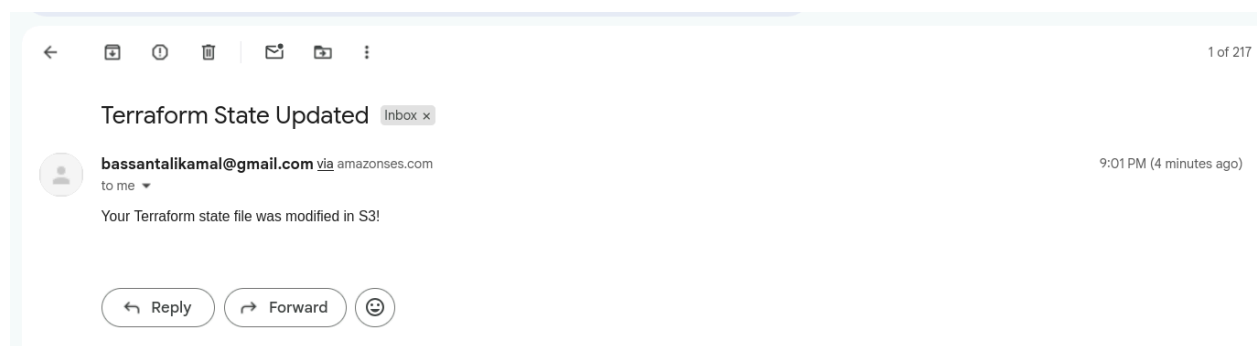
You can attach up to 10 managed policies.

Search Filter by Type All types < 1 > [Settings](#)

☐	Policy name i2	Type	Attached entities
☐	AWSLambdaBasicExecutionRole	AWS managed	1
☐	lambda-ses-policy	Customer inline	0



9-create trigger to detect changes in state file and send the email



10-create rds (mysql)

app-mysql-db

app-mysql-db

Modify Actions

Summary

DB identifier

app-mysql-db

CPU

32.95%

Status

Available

Class

db.t3.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

eu-north-1a

Recommendations

Security group rules (2)

Filter by Security group rules

Security group	Type	Rule
rds-sg (sg-04116feb2497c0239)	EC2 Security Group - Inbound	sg-041767585854f938b
rds-sg (sg-04116feb2497c0239)	CIDR/IP - Outbound	0.0.0.0/0

sg-041767585854f938b - app-sg

sg-041767585854f938b - app-sg

Actions

Details

Security group name

app-sg

Security group ID

sg-041767585854f938b

Description

Managed by Terraform

VPC ID

[vpc-0304dd242baa84924](#)

Owner

707938860152

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-0d1e2edc057a4b671	IPv4	Custom TCP	TCP	3000	172.16.0.0/16
☐	-	sgr-0e7a11cd8415ab777	IPv4	SSH	TCP	22	172.16.0.0/16

11-create elastic cache (redis)

Amazon ElastiCache announces Valkey starting as low as \$6/month

Create Valkey cache

Introducing Valkey, an open source project that is a drop in replacement of Redis OSS, now available on ElastiCache, with up to 33% lower price than other engines. [Learn more](#)

Redis OSS caches (1) Info

Upgrade to Valkey View details Actions Create cache

Find Redis caches

Cache name	Status	Description	Engine version	Configuration	Created
☐ app-redis	Available	-	7.1.0	cache.t3.micro	May 6, 2024